

Constructing Masterpieces From Missing Pieces



Invited Speaker

Huiyu Zhou

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Zoom Meeting: 801 137 0362

Biography

Huiyu Zhou received a Bachelor of Engineering degree in Radio Technology from Huazhong University of Science and Technology of China, and a Master of Science degree in Biomedical Engineering from University of Dundee of United Kingdom, respectively. He was awarded a Doctor of Philosophy degree in Computer Vision from Heriot-Watt University, Edinburgh, United Kingdom. Dr. Zhou currently is a full Professor at School of Computing and Mathematical Sciences, University of Leicester, United Kingdom. He has published over 600 peer-reviewed papers in the field. His research work has been or is being supported by UK EPSRC, ESRC, AHRC, MRC, EU, Innovate UK, Royal Society, British Heart Foundation, Leverhulme Trust, Puffin Trust, Alzheimer's Research UK, Invest NI and industry. Homepage: <https://le.ac.uk/people/huiyu-zhou>

Abstract

Image completion is a challenging task, particularly when ensuring that generated content seamlessly integrates with existing parts of an image. While recent diffusion models have shown promise, they often struggle with maintaining coherence between known and unknown (missing) regions. This issue arises from the lack of explicit spatial and semantic alignment during the diffusion process, resulting in content that does not smoothly integrate with the original image. Additionally, diffusion models typically rely on global learned distributions rather than localized features, leading to inconsistencies between the generated and existing image parts. In this work, we propose ConFill, a novel framework that introduces a Context-Adaptive Discrepancy (CAD) model to ensure that intermediate distributions of known and unknown regions are closely aligned throughout the diffusion process. By incorporating CAD, our model progressively reduces discrepancies between generated and original images at each diffusion step, leading to contextually aligned completion. Moreover, ConFill uses a new Dynamic Sampling mechanism that adaptively increases the sampling rate in regions with high reconstruction complexity. This approach enables precise adjustments, enhancing detail and integration in restored areas. Extensive experiments demonstrate that ConFill outperforms current methods, setting a new benchmark in image completion.